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Course: Big Data Individual Assignment Report

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Topic: Exploring Environmental and Road Factors Influencing Traffic Collision Severity

Introduction

Road safety remains a major concern in large urban areas where traffic density and complex road networks increase the likelihood of collisions. Understanding the factors associated with severe road accidents can help policymakers and city planners design safer transportation systems. This project analyzes the Motor Vehicle Collisions with KSI (Killed or Seriously Injured) dataset obtained from the Toronto Open Data Portal. The dataset contains detailed records of traffic collisions reported within the City of Toronto, including information about collision severity, lighting conditions, road surface conditions, impact types, and other environmental factors. The objective of this project is to explore patterns in collision data and identify factors that may influence accident severity. Apache Spark was used to process and analyze the dataset efficiently, Spark Structured Streaming was used to simulate how collision data could be processed in a streaming environment, and MongoDB Atlas was used as a NoSQL storage layer to persist all analytical results across named collections.

Tools and Architecture Used: The project was implemented using a big data processing workflow built on Apache Spark. The main tools and technologies used include:

- **Apache Spark (PySpark):** Used for data ingestion, cleaning, transformations, aggregations, and streaming simulation.
- **Spark Structured Streaming:** Used to simulate real-time data ingestion by processing collision data incrementally from Parquet files.
- **Python (PySpark API):** Used to write Spark transformations and perform analytical computations.
- **Matplotlib:** Used to create visualizations such as bar charts and line graphs to present insights.
- **Parquet Storage Format:** Used for storing intermediate and streaming output data because it provides efficient column-based storage.
- **MongoDB Atlas (PyMongo):** Used as the second Big Data tool (distributed NoSQL storage) to persist all eight analytical result sets as named collections, enabling structured querying of processed results beyond flat file storage.

The overall data processing pipeline for the project follows this structure:

- Data Source (Toronto Open Data Portal)
- Data Ingestion using Apache Spark
- Data Cleaning and Preprocessing
- Feature Engineering (year, month, day_of_week, hour_of_day)
- Spark Transformations, Aggregations and Visualization
- Streaming Simulation using Spark Structured Streaming
- MongoDB Atlas Integration (NoSQL persistent storage)
- Storage of Results (Parquet local backup + MongoDB collections)

The dataset used in this project is Motor Vehicle Collisions with KSI Data, which records collisions in Toronto that resulted in serious injuries or fatalities. The data is collected from the Toronto Police Service and includes collision records dating from 2006 to 2025. The dataset contains several attributes describing collision circumstances, including:

- Date and time of the collision (accdate)
- Street names and intersection details (stname1, stname2)
- Geographic coordinates (latitude, longitude)
- Collision severity classification (acclass)
- Type of collision impact (impactype)
- Lighting conditions (light)
- Road surface conditions (rdsfcond)
- Road classification (road_class)
- Neighbourhood and ward information (neighbourhood, wardname)
- Vehicle and participant information

The dataset originally contained 20,419 records and more than forty variables describing various aspects of each collision event.

Data Cleaning and Preprocessing

- Irrelevant columns (stname3, fatal_no, pedact, pedcond, cyclistype, cycact, cyccond, drivact, drivcond) were dropped due to high missing values or irrelevance to the core analysis objectives.

- Rows with missing values in key analytical columns (accdate, acclass, light, rdsfcond, impactype, road_class) were dropped, ensuring data integrity for primary analyses. The DataFrame size was reduced from 20419 to 19718 rows.
- Duplicate rows were checked and found to be non-existent.

```
# drop irrelevant columns

columns_to_drop = [
    "stname3",
    "fatal_no",
    "pedact",
    "pedcond",
    "cyclistype",
    "cycact",
    "cyccond",
    "drivact",
    "drivcond"
]

df = df.drop(*columns_to_drop)
```

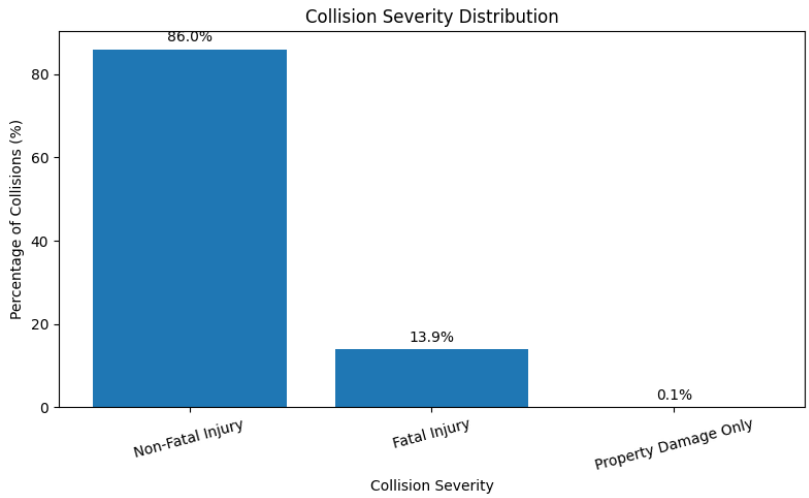
```
# drop rows missing key analytical variables

columns_for_na_drop = [
    "accdate",
    "acclass",
    "light",
    "rdsfcond",
    "impactype",
    "road_class"
]

df = df.na.drop(subset=columns_for_na_drop)
```

Feature Engineering: Additional features (year, month, day_of_week, hour_of_day) were derived from the accdate column and converted to a timestamp to allow the dataset to be analyzed across different time periods.

Spark Transformations and Aggregations

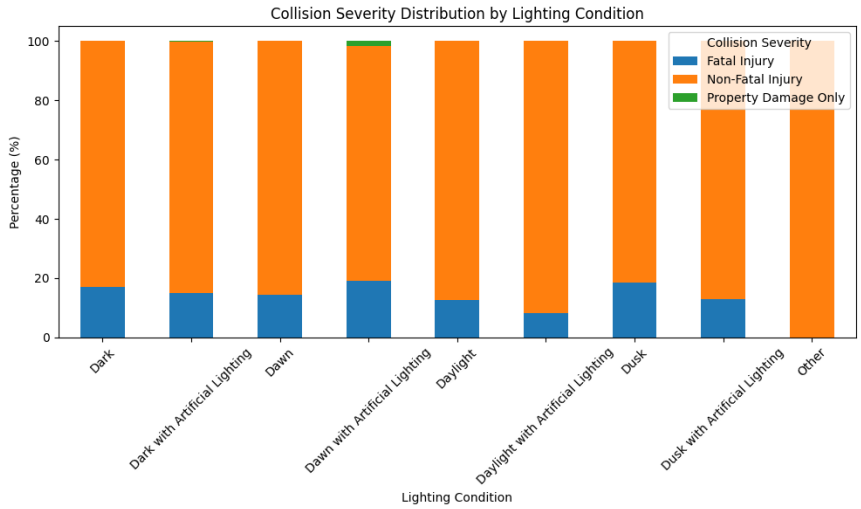
Apache Spark transformations and aggregations were applied to analyze patterns within the collision dataset. Using Spark DataFrame operations such as groupBy, count, and window functions, key metrics were computed across several variables including collision severity, lighting conditions, road surface conditions, impact types, temporal trends, high-risk neighbourhoods, and time-of-day severity patterns.

I. **Collision Severity Distribution:** To understand the overall distribution of collision outcomes, the dataset was grouped by the collision severity classification (acclass).



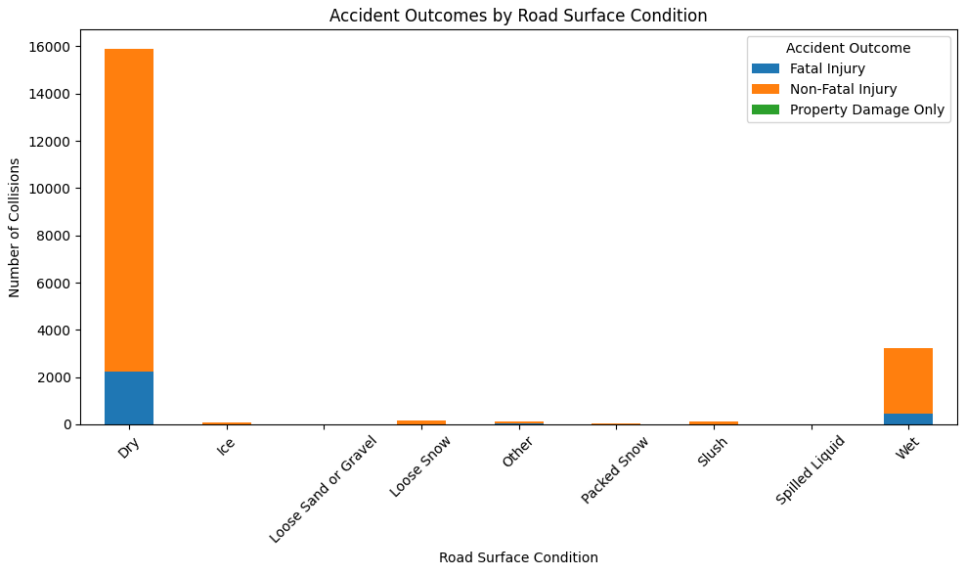
- **Finding:** Non-Fatal Injuries constituted the vast majority (85.99%) of collisions, Fatal Injuries accounted for 13.92%, and Property Damage Only cases were minimal (0.09%).
- **Insight:** The dataset predominantly captures less severe injury-related collisions, indicating the focus on KSI data.

II. **Lighting Conditions and Collision Severity:** To examine how environmental visibility conditions influence accident outcomes, the dataset was grouped by lighting condition (light) and collision severity (acclass).



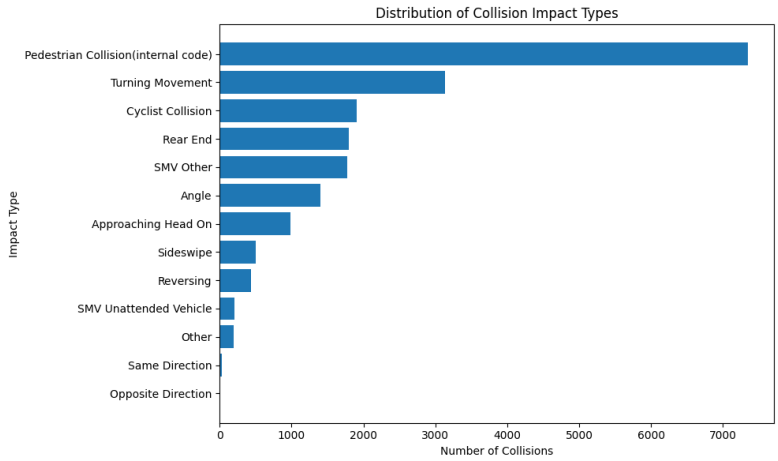
- **Finding:** Periods of reduced natural light (Dusk, Dawn, Dark) showed a higher proportion of fatal injuries (ranging from 12.93% to 19.13%) compared to Daylight conditions (12.53% for Daylight and 8.21% for Daylight with Artificial Lighting).
- **Insight:** Reduced visibility during low natural light conditions likely contributes to increased collision severity.

III. **Road Surface Conditions and Accident Outcomes:** To explore how road conditions influence accident outcomes, collisions were grouped by road surface condition (rdsfcond) and collision severity (acclass).



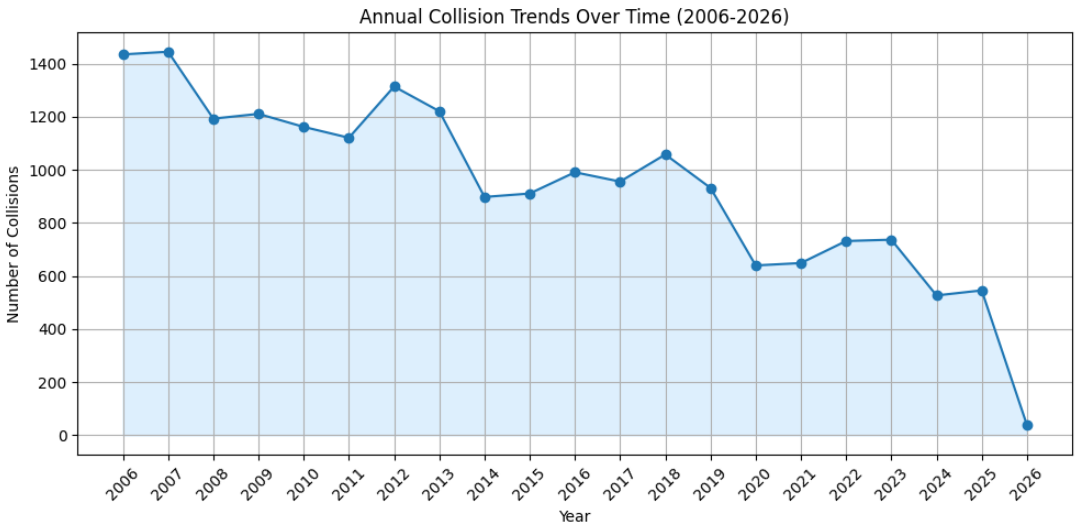
- **Finding:** While Dry (80.65%) and Wet (16.46%) conditions accounted for most collisions, Other road surface conditions (29.93%) and Packed Snow (25.0%) had significantly higher proportions of fatal injuries.
- **Insight:** Less common but highly hazardous road conditions (Other, Packed Snow) carry a disproportionately higher risk of fatal outcomes, warranting specific investigation.

Collision Impact Type Analysis: To identify the most common types of collisions, the dataset was grouped by impact type (impactype).



- **Finding:** Pedestrian Collision (internal code) (18.22%) and Approaching Head On' (16.99%) exhibited the highest proportions of fatal injuries. 'Reversing' also had a notable fatal injury proportion (16.28%).
- **Insight:** These impact types are associated with greater severity, suggesting targets for prevention efforts and public awareness campaigns.

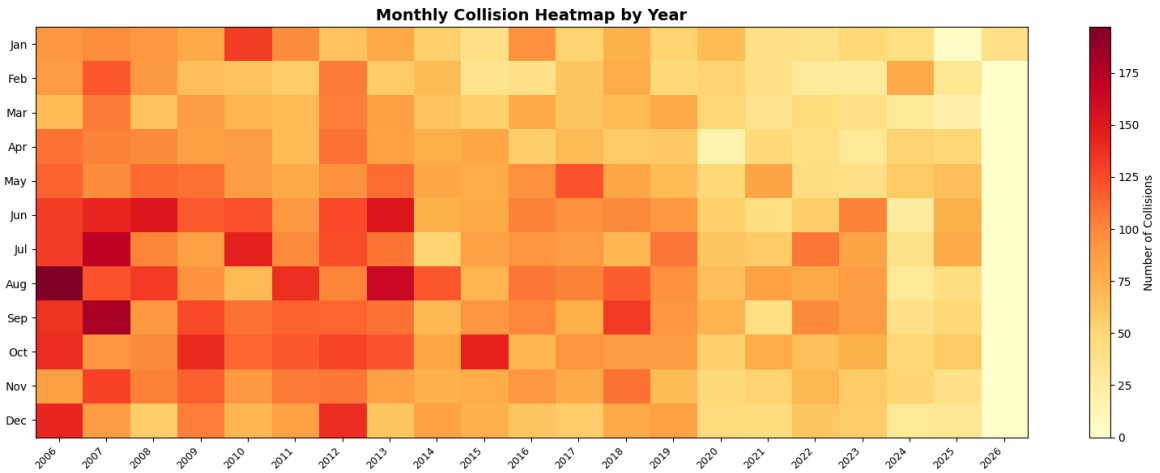
IV. **Collision Trends Over Time:** To examine how collision occurrences change over time, the dataset was aggregated by year.



- **Finding:** The number of collisions showed yearly fluctuations and clear monthly seasonality. Peak months for collisions were generally August and June, while February and December often had lower counts.

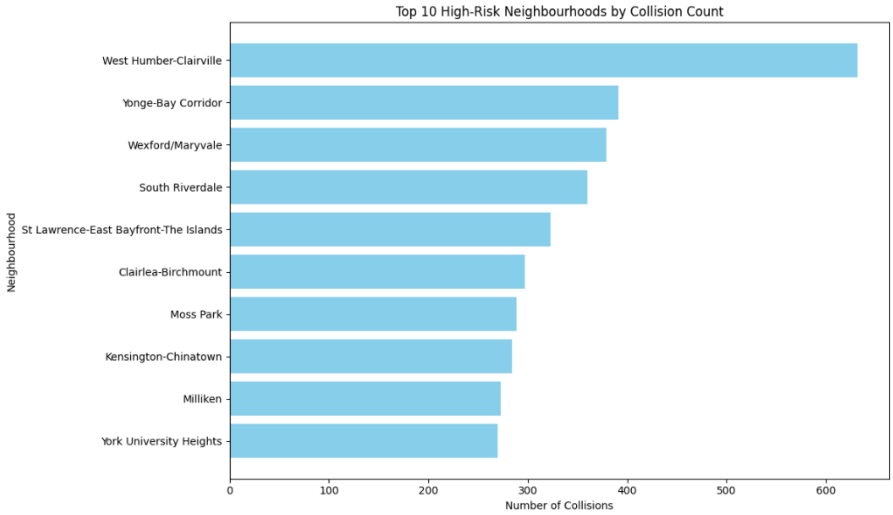
- **Insight:** Seasonal variations likely correlate with factors like weather, driving conditions, and human activity patterns.

V. **Monthly Collision Heatmap:** To identify seasonal patterns across all years simultaneously, collision counts were aggregated by both month and year and visualized as a heatmap matrix.



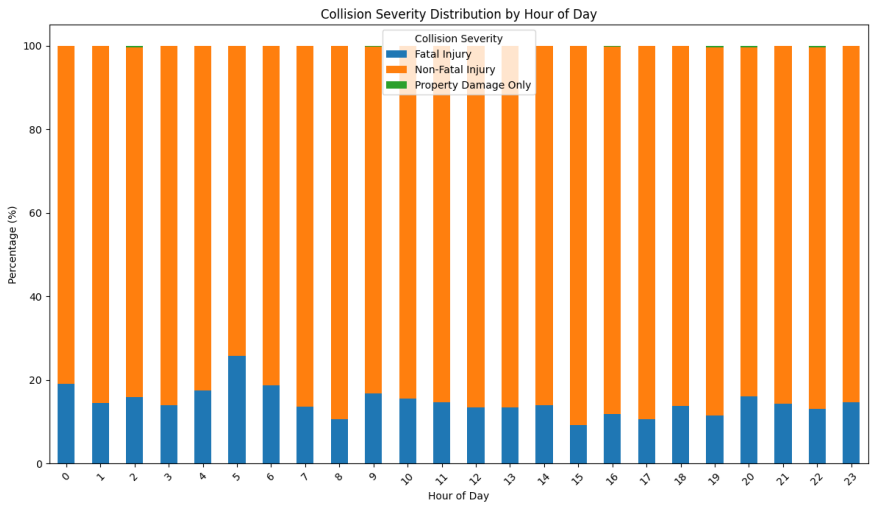
- **Finding:** Summer months (June–August) consistently showed elevated collision counts across years. A visible decline in collision volume is observable from 2020 onwards, coinciding with reduced traffic during the COVID-19 pandemic period, with partial recovery visible from 2022.
- **Insight:** The heatmap reveals multi-year seasonal patterns more clearly than a single yearly trend line, making it a more powerful tool for identifying persistent high-risk periods.

VI. **High-Risk Neighbourhood Analysis:** To identify spatial patterns, the dataset was grouped by neighbourhood and the top 10 neighbourhoods by total collision count were ranked and visualized.



- **Finding:** Several Toronto neighbourhoods showed disproportionately high collision concentrations, pointing to specific urban zones where road safety interventions would have the greatest impact.
- **Insight:** The spatial analysis complements the temporal findings and provides location-based evidence to support targeted infrastructure and enforcement policies.

VII. **Time-of-Day Severity Analysis:** To examine how collision severity varies throughout the day, an hour_of_day feature was extracted and cross-tabulated with collision severity using a window function partitioned by hour.



- **Finding:** Late night and early morning hours showed a higher proportion of fatal injuries compared to daytime hours, consistent with the lighting conditions finding. Peak collision volumes occurred during afternoon rush hours (15:00-18:00), though their fatal proportion was comparatively lower.
- **Insight:** Volume and severity are inversely distributed across the day, reinforcing that nighttime collisions, despite being fewer, carry a disproportionately higher risk of fatality.

Streaming Simulation Using Spark Structured Streaming

Objective: To simulate live traffic collision updates and process micro-batches continuously.

Implementation:

I. **Incremental Source Preparation:** Incremental Source Preparation: The historical df was split into yearly subsets and written incrementally to a directory. This simulated the arrival of new data over time.

II. **Streaming Ingestion & Aggregation:** Spark Structured Streaming was configured to read Parquet files from this directory. As new files were detected, Spark processed them in micro-batches.

During each batch, a simple aggregation operation (count(*)) was performed to track the total number of records being processed.

III. **Persistent Storage:** The streaming query was configured to write the aggregated counts from each micro-batch to a persistent Parquet sink using the foreachBatch output mode. This approach correctly handles global aggregations by writing the complete, updated state from each batch.

```
Writing data for 21 years into /tmp/collision_stream_source_incremental...
Written data for year 2006. Total records: 1435
Written data for year 2007. Total records: 1445
Written data for year 2008. Total records: 1193
Written data for year 2009. Total records: 1211
Written data for year 2010. Total records: 1162
Written data for year 2011. Total records: 1121
Written data for year 2012. Total records: 1314
Written data for year 2013. Total records: 1221
Written data for year 2014. Total records: 898
Written data for year 2015. Total records: 911
Written data for year 2016. Total records: 991
Written data for year 2017. Total records: 956
Written data for year 2018. Total records: 1058
Written data for year 2019. Total records: 932
Written data for year 2020. Total records: 640
Written data for year 2021. Total records: 649
Written data for year 2022. Total records: 732
Written data for year 2023. Total records: 737
Written data for year 2024. Total records: 527
Written data for year 2025. Total records: 546
Written data for year 2026. Total records: 39
Streaming source prepared.
```

Execution Confirmation: The streaming query successfully started and, after processing all yearly batches, reported a total count of 19718 records. This demonstrated that the streaming pipeline is functional and ready for live data integration.

```
Current counts from streaming_counts_per_batch:
+-----+
|count|
+-----+
|19718|
+-----+

Spark Structured Streaming query stopped.
```

MongoDB Integration

Implementation:

I. **Connection and Security:** A MongoDB Atlas free-tier cluster was created for this project. The database credentials were stored securely using Google Colab's user data Secrets panel instead of being hardcoded in the notebook.

II. **Data Storage:** All eight analytical Spark DataFrames were converted to Pandas DataFrames and then inserted as document collections into the collision_analysis MongoDB database. Before inserting the new data, each collection was cleared to avoid duplicates and ensure the process could be repeated safely. Collections stored include: collision_severity_distribution, lighting_conditions_analysis, road_surface_conditions_analysis, impact_type_analysis, collision_trends_over_time, monthly_yearly_collision_heatmap_data, high_risk_neighbourhoods, and time_of_day_severity_analysis.

III. **Verification:** After inserting the data, a sample document was retrieved from each collection to verify that the data had been stored correctly. The MongoDB connection was then closed in a finally block to ensure proper resource management. The number of inserted documents was printed for all eight collections, confirming that the integration worked successfully.

```
Successfully connected to MongoDB database 'collision_analysis'

Processing collection: collision_severity_distribution
Successfully saved 3 records to collection 'collision_severity_distribution'
Sample data from MongoDB for this collection:
{'_id': ObjectId('69b6d13793fef89a3e1ebf4d'), 'acclass': 'Non-Fatal Injury', 'count': 16956, 'proportion_percent': 85.99}

Processing collection: lighting_conditions_analysis
Successfully saved 20 records to collection 'lighting_conditions_analysis'
Sample data from MongoDB for this collection:
{'_id': ObjectId('69b6d13793fef89a3e1ebf50'), 'light': 'Dark', 'acclass': 'Non-Fatal Injury', 'count': 3002, 'total_light_count': 3617, 'proportion_percent': 83.0}

Processing collection: road_surface_conditions_analysis
Successfully saved 18 records to collection 'road_surface_conditions_analysis'
Sample data from MongoDB for this collection:
{'_id': ObjectId('69b6d13893fef89a3e1ebf64'), 'rdsfcond': 'Dry', 'acclass': 'Non-Fatal Injury', 'count': 13681, 'total_rdsfcond_count': 15913, 'proportion_percent': '85.97%'}

Processing collection: impact_type_analysis
Successfully saved 31 records to collection 'impact_type_analysis'
Sample data from MongoDB for this collection:
{'_id': ObjectId('69b6d13993fef89a3e1ebf76'), 'impactype': 'Angle', 'acclass': 'Non-Fatal Injury', 'count': 1224, 'total_impactype_count': 1405, 'proportion_percent': 87.12}

Processing collection: collision_trends_over_time
Successfully saved 21 records to collection 'collision_trends_over_time'
Sample data from MongoDB for this collection:
{'_id': ObjectId('69b6d13993fef89a3e1ebf95'), 'year': 2006, 'count': 1435}

Processing collection: monthly_yearly_collision_heatmap_data
Successfully saved 241 records to collection 'monthly_yearly_collision_heatmap_data'
Sample data from MongoDB for this collection:
{'_id': ObjectId('69b6d13a93fef89a3e1ebfaa'), 'year': 2006, 'month': 1, 'count': 92}

Processing collection: high_risk_neighbourhoods
Successfully saved 159 records to collection 'high_risk_neighbourhoods'
Sample data from MongoDB for this collection:
{'_id': ObjectId('69b6d13c93fef89a3e1ec09b'), 'neighbourhood': 'West Humber-Clairville', 'count': 632}
```

Output and Visualization

Storage: All analytical results were successfully stored both as Parquet files (local backup) and as named collections in MongoDB Atlas (NoSQL persistent storage). Eight collections were written to the collision_analysis database, covering all analytical dimensions from severity distribution to neighbourhood risk and time-of-day patterns.

Visualizations: Visualizations (bar charts, stacked bar charts, line plots, a monthly heatmap, and a horizontal bar chart for neighbourhoods) were generated using matplotlib and pandas for each analytical task to illustrate patterns in collision severity, environmental factors, impact types, temporal trends, spatial risk distribution, and time-of-day severity profiles.

Conclusion

This project successfully demonstrated a complete mini Big Data workflow using Apache Spark, Spark Structured Streaming, and MongoDB Atlas. The batch processing pipeline extracted deep insights from 19,718 historical collision records across eight analytical dimensions: collision severity, lighting conditions, road surface conditions, impact types, yearly and monthly temporal trends, high-risk neighbourhood spatial analysis, and time-of-day severity profiling. MongoDB Atlas served as the second Big Data tool, providing a distributed NoSQL storage layer where all results were persisted as queryable document collections. Spark Structured Streaming simulated live data ingestion by processing yearly batches incrementally.